Math 372: Practice Midterm #1

Problem 1. Determine whether the following statements are true or false. (Circle **T** or **F**):

(a) **T** | **F** If two events are independent then they cannot both happen.

(b) **T** | **F** If $E_1, \dots, E_n$ are pairwise disjoint events each with probability at least $1/4$, then $n \leq 4$.

(c) **T** | **F** If $A \subseteq B$ then $\mathbb{P}[A \mid B] = \mathbb{P}[A]$

(d) **T** | **F** If $\mathbb{P}[A] = .6$ and $\mathbb{P}[B] = .6$, then $\mathbb{P}[A \cap B] \geq .2$

(e) **T** | **F** If $\mathbb{P}[A^c \cap B^c] = 0$ then $\mathbb{P}[A \cup B] = 1$

(f) **T** | **F** If A and B are events, then $\mathbb{P}[B] = \mathbb{P}[B \mid A] + \mathbb{P}[B \mid A^c]$

(g) **T** | **F** For any events A and B, $(A \cap B^c)^c = A^c \cup B$

Problem 2 (set theory)**.** Let A be the event that a randomly-selected person drinks coffee regularly. Let B be the event that they drink soda regularly. Suppose that 45% of all people regularly consume coffee, 35% regularly consume soda, and 60% regularly consume at least one of these two drinks. *(Hint: draw a Venn diagram.)*

(a.) What is the probability that a randomly-selected person regularly consumes both coffee and soda?

(b.) What is the probability that a randomly-selected person doesn't regularly consume at least one of these drinks?

(c.) Find $\mathbb{P}[A \backslash B]$

(d.) Find $\mathbb{P}[B \mid A \cup B]$

(e.) Are A and B independent? Justify your answer.

(f.) Are A and B disjoint? Justify your answer

Problem 3 (combinatorics)**.** A dessert company has 31 distinct desserts on its menu: 5 types of pie, 9 types of cake, and 17 flavors of ice cream. *(You don't need to simplify your answers for this problem)*

(a) After purchasing 4 different desserts, a customer decides he will eat them all in a row, and he cares about the order in which he eats them. How many different ways can he eat his 4 desserts?

(b) Suppose 7 different desserts are selected from the menu. How many ways are there to do this if (i) we care about the order in which the 7 deserts are selected, and (ii) if we don't care about the order?

(c) If 9 desserts are randomly selected, how many ways are there to obtain 3 deserts of each type (pie, cake, and ice cream)?

(d) If 5 desserts are randomly selected, what's the probability that they are all of the same type (i.e., all pie, all cake, or all ice cream)?

Problem 4 (discrete r.v.)**.** Let X and Y denote two rolls of six-sided dice. Let $W = |X - Y|$.

(a) What values can W take?

(b) Display the pmf of W in a suitable table.

(c) What is $\mathbb{E}[W]$?

(d) Find $\mathbb{P}[W > 2]$

(e) Find $\mathbb{P}[W > 2 \mid X = 6]$

Problem 5 (conditional probability)**.** An email provider observes that 80% of emails are spam. It intends to filter spam emails before they reach the inbox. The filter's accuracy for detecting spam email is 98% and chances of tagging a non-spam email as spam is 4%.

(a) Draw a tree diagram to illustrate this situation

(b) If a certain email is tagged as spam, find the probability that it is not a spam email. That is, find

$$\mathbb{P}\left[\text{email is not spam} \mid \text{email is tagged as spam}\right]$$

Problem 6 (set theory, conditional probability)**.** Consider randomly selecting a student at a large university. Let A be the event that the selected student has a Visa card, let B be the event that the student has a MasterCard, and let C be the event that they have an American Express card. Suppose that

$$\mathbb{P}[A] = .6, \quad \mathbb{P}[B] = .4, \quad \mathbb{P}[C] = .2$$
$$\mathbb{P}[A \cap B] = .3, \quad \mathbb{P}[A \cap C] = .15, \quad \mathbb{P}[B \cap C] = .1,$$
$$\text{and} \quad \mathbb{P}[A \cap B \cap C] = .08.$$

(a) What is the probability that the selected student has at least one of the three types of cards?

(b) What is the probability that the selected student has both a Visa card and a MasterCard, but not an American Express card?

(c) Calculate and interpret $\mathbb{P}[B \mid A]$ and $\mathbb{P}[A \mid B]$.

(d) If we learn that the selected student has an American Express card, what is the probability that she or he also has both a Visa card and a MasterCard?

(e) Given that the selected student has an American Express card, what is the probability that she or he has at leaset one of the other two types of cards?

Math 372: Practice Midterm #2

Problem 1 (combinatorics)**.** A furniture consignment store contains 24 pieces of furniture: 9 dressers, 5 tables, and 10 chairs. Suppose that three pieces of furniture are randomly selected to be put in a display showroom.

(a) What is the probability that the showroom has exactly two chairs?

(b) What is the probability that the showroom contains all furniture of the same type (i.e., all dressers, all tables, or all chairs)?

(c) What is the probability that the showroom contains one piece of furniture of each type?

(d) Suppose pieces of furniture are sold at random, one at a time. What is the probability that it takes at least 6 sales before the first table is sold?

Problem 2 (conditional probability)**.** Two cards are randomly drawn from a deck of cards. Let A be the event that at least one ace is drawn. Let A_{s} be the event that the ace of spades is chosen. And let B be the event that both cards are aces. Compute the following conditional probabilities:

(a) $\mathbb{P}[B \mid A_{\mathrm{s}}]$

(b) $\mathbb{P}[B \mid A]$

(c) $\mathbb{P}[A_{\mathrm{s}} \mid B]$

Problem 3 (set theory, conditional probability)**.** A sample of 2026 students were asked about their eye color, with the following results:

	Blue	**Brown**	**Green**	**Hazel**	**Total**
Male	370	352	198	187	1107
Female	359	290	110	160	919
Total	729	642	308	347	2026

Suppose that one of these 2026 students is randomly selected. Let F denote the event that the selected individual is female, and A, B, C and D represent the events that they have blue, brown, green, and hazel eyes, respectively.

(a) Calculate $\mathbb{P}[F]$ and $\mathbb{P}[C]$

(b) Calculate $\mathbb{P}[F \cap C]$. Are these events independent? Why or why not?

(c) If the selected individual has green eyes, what's the probability that he or she is a female?

(d) If the selected individual is female, what is the probability that she has green eyes?

(e) What is the "conditional distribution" of eye color for females? (i.e., $\mathbb{P}[A|F], \mathbb{P}[B|F], \mathbb{P}[C|F], \mathbb{P}[D|F]$), and what is it for males? Compare the two distributions.

Problem 4 (combinatorics, conditional probability)**.** A coin is flipped 10 times.

(a) What is the probability that there are at least 7 heads? Don't simplify your answer.

(b) You recieve credible information that there were at least 5 heads in the 10 coin tosses. Given that new information, what is the probability that there were at least 7 heads? Don't simplify your answer.

Problem 5 (discrete r.v.)**.** Consider a deck of four cards consisting of one card numbered '1', two cards numbered '2', and one card numbered '3'. The deck is suffled thoroughly. Let D denote the difference between the numbers on the top two cards, ignoring the sign.

(a) Display the probability mass function of D in a suitable table.

(b) Find $\mathbb{E}[D]$

(c) Find $\mathrm{Var}(D)$